

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1705

Cheboygan 138 kV
May 19, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1705. The Interconnection Customer proposes a 207.8 MW solar photovoltaic facility interconnecting at the Cheboygan 138 kV substation in the Virginia area. The request was received on March 28, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 567 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,562,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Cheboygan Substation / Mecklenburg Power Sta Substation 138 kV line	Person / Mecklenburg Power Station 345 kV	116.5	433	59.5
Mecklenburg Power Station / Bear 345 kV	Bear / Clover 345 kV	102.5	3,288	35.4
Fluvanna P.S / Bear 345 kV	Birchwood / Leavitt 345 kV	100.0	2,071	21.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cheboygan Substation / Mecklenburg Power Sta Substation 138 kV line	Rebuild line with higher-capacity conductor	\$5,773,000	55% (\$3,200,000)
Mecklenburg Power Station Substation / Bear 345 kV Line	Reconductor line	\$56,693,000	4% (\$2,131,000)
Fluvanna P.S / Bear 345 kV	Reconductor line	\$52,734,000	2% (\$1,231,000)
	Total allocated to project		\$6,562,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.